
Verify-or-Abstain Evidence Pipelines: When Cryptographic Custody Is Necessary but Not Sufficient

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Abstract

1 Scientific evidence pipelines increasingly rely on content-addressed custody and
2 hash-valid artifacts, yet syntactically valid objects can remain semantically in-
3 consistent. We present an evidence-pipeline reliability study centered on explicit
4 terminal states, row-level accounting invariants, and closed-vocabulary identity
5 guards. Audits, deterministic ablations, and retrospective successor evaluations
6 show that (i) a historical aggregate reported zero abstentions while row-level data
7 contained three; (ii) a successor implementation with derived aggregates restores
8 invariant consistency; (iii) a deterministic identity contract passes under synthetic
9 fixtures but a single observed-source pair admits downstream passage when a
10 guard is bypassed; and (iv) the historical G1/G2 corpora were not recovered from
11 admitted provenance, while contemporary successor evaluations were run on ad-
12 mitted subsets. The morphology null benchmark was independently reproduced
13 and retained as negative evidence. We do not claim therapeutic efficacy, clinical
14 utility, biological rescue, population-level biomedical error rates, treatment rec-
15 ommendations, or real-world evidence.

16 1 Introduction

17 Machine-learning systems for biomedical hypothesis generation must separate *what was computed*
18 from *what may be claimed*. Content-addressed storage and hash verification establish integrity of
19 bytes, not meaning: aggregates can disagree with row ledgers, evaluation wiring can fail to exercise
20 intended disagreements, and missing corpora can leave historical numbers unverified. We frame this
21 work as an evidence-pipeline control problem: normalize, validate, admit or abstain, derive results,
22 and attach claim ceilings before projection to manuscripts [8, 5].

23 2 Related Work and Terminology

24 Selective prediction and reject options allow models to abstain under uncertainty [2, 3]. Workflow
25 and data provenance standards document lineage without guaranteeing semantic correctness [8].
26 Biomedical identifier normalization relies on authorities such as HGNC and UniProt [1, 6]. Re-
27 producibility guidance distinguishes replication from reproducibility and encourages reporting null
28 results [4]. Regulatory Real-World Evidence frameworks define boundaries we do not claim here
29 [7].

30 We distinguish *cryptographic custody* (hash-valid, content-addressed storage) from *semantic verifi-*
31 *cation* (terminal-state invariants, identity resolution, aggregate-row consistency).

32 **Synthetic fixture.** An intentionally constructed input used to test deterministic software behavior.

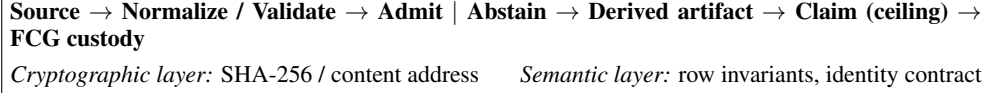


Figure 1: Verify-or-abstain evidence pipeline. Custody without semantic checks can admit inconsistent aggregates.

Experiment	Unit	Guard	Condition	N	Ceiling
GAP accounting repair	row	aggregate invariant	successor run	4	semantic audit
G3 contract	fixture	identity	Lane A	synth.	deterministic
G3 source ablation	pair	identity bypass	Lane B	1	descriptive
G1 successor eval	record	sequence guard	contemporary	364	retrospective
G2 successor eval	record	CNV/accounting	contemporary	746	retrospective
G1 historical prov.	corpus	—	hist. not recovered	—	audit only
G2 historical prov.	corpus	—	hist. not recovered	—	audit only
Morphology null	ranking	shuffle null	reproduced	50	repurposing hyp.

Table 1: Primary results from frozen experiment artifacts. Historical G1/G2 counts are excluded.

Canonical contract fixture. A project-defined synthetic or curated fixture whose expected output is fixed before execution and used to test a deterministic contract.

Observed source record. An actual record retrieved from an identified external source rather than invented for testing. Evaluations using such records are *observed-source evaluations*; they are not Real-World Evidence (RWE) unless regulatory RWD/RWE criteria are satisfied [7].

External biological evidence. A source-backed biological record or publication used as evidence for a biological assertion.

Abstention / reject option. Following Chow [2] and selective-prediction literature [3], a pipeline may emit a terminal ABSTAIN state rather than admit an under-supported record.

Closed-vocabulary identity resolution. Map source strings to a frozen alias table with exact, mapped, or unresolved outcomes; unresolved inputs must not silently pass as canonical.

Provenance. Origin and processing lineage of entities [8]. Hash equality does not imply semantic consistency between derived artifacts.

3 Verify-or-Abstain Architecture

Figure 1 sketches the pipeline: SOURCE → normalize/validate → admit or abstain → derived result → claim with ceiling → frozen custody graph. Semantic verification (terminal states, invariants, identity guards) complements cryptographic custody.

4 Experimental Design

Experiments are preregistered where noted; August 13 historical runs are preserved and not retroactively relabeled. EXP-GAP-ACCOUNTING-001.1 repairs aggregate derivation from rows. EXP-004 ablates a closed-vocabulary identity guard (Lane A: contract; Lane B: $N=1$ observed-source pair). EXP-002/003 provenance audits and external recovery (EXP-002-PROV.1, EXP-003-PROV.1) search admitted custody without rerunning frozen closeouts; contemporary successor evaluations (EXP-002-SUCCESSOR-001, EXP-003-SUCCESSOR-001) run on admitted subsets. EXP-005 preregistration remains locked but corpus/metric incomplete (not executed). EXP-006 independently reproduces a morphology null benchmark.

5 Results

Table 1 summarizes terminal classifications and claim ceilings from machine-readable receipts.

61 **GAP accounting.** Historical summary reported zero abstentions; row-level candidates contained
62 three ABSTAIN terminals. Successor run: $N_{\text{input}}=4$, $N_{\text{admitted}}=1$, $N_{\text{abstained}}=3$, summary abstain
63 count = 3; invariants pass.

64 **Identity guard.** Lane A: deterministic closed-vocabulary contract passes. Lane B: one ob-
65 served source/canonical pair (TAZ/cardioliipin module $\rightarrow$ TAZ) yields ABSTAIN under guard
66 and downstream admission under bypass (*unsupported identity admission*); not a population error
67 rate. Historical evaluation invoked `reconcile(symbol, symbol)`; regression confirms the wiring
68 defect.

69 **G1 successor evaluation.** On the admitted contemporary ClinVar subset ($N=364$), a retrospective
70 successor sequence guard emitted 98 records and abstained 266 ($364=98+266$): frameshift 131, no
71 protein notation 118, residue mismatch 16, position out of range 1. These abstention categories are
72 accounting terminals in the contemporary successor evaluation, not a population biomedical error
73 rate and not proof that source records are globally incorrect. This is *not* exact historical reproduction.

74 **G2 successor evaluation.** Contemporary fetch accounting: 746 IDs fetched, 364 kept, 382 ex-
75 cluded as large/multi-gene copy-number events, 0 without summary ($746=364+382+0$). The 382
76 exclusions reflect unsupported gene-specific attribution under the frozen G2 predicate, not a sep-
77 arately validated “silent error” rate. A historical 100-ID ESummary failure (~ 16.5 MB XML vs.
78 10 MB JSON ceiling) is recovered via split-and-retry; the contemporary 746-row corpus does not
79 reproduce a historical 742-row corpus.

80 **Provenance recovery.** Admitted external bootstrap package (206 MB, SHA-256 verified) does not
81 contain recoverable historical G1/G2 derivation chains; numeric substring matches in unrelated CSV
82 fields are rejected as evidence.

83 **Null benchmark.** The morphology null benchmark was independently reproduced: known-pair
84 rank 28/50, reciprocal-rank enrichment vs. shuffle < 1 ; the negative result was retained and not
85 suppressed.

86 6 Discussion and Limitations

87 Scope is intentionally small: $N=1$ observed-source ablation; the historical G1/G2 corpora were not
88 recovered from admitted provenance (contemporary successor subsets were evaluated separately);
89 same designers performed parts of evaluation; no therapeutic efficacy, biological rescue, clinical util-
90 ity, treatment recommendations, observed-source evaluations framed as RWE, or population preva-
91 lence claims; EXP-005 replication not executed pending corpus freeze; infrastructure integrity does
92 not establish semantic correctness. No structure-prediction execution is part of admitted experiment
93 evidence.

94 7 Reproducibility and Provenance

95 Artifacts, preregistrations, and SHA-256 manifests reside in the project Frozen Custody Graph.
96 Anonymous supplementary materials omit identifying metadata. Historical unverified counts remain
97 quarantined as UNVERIFIED_HISTORICAL_CLAIM. The submitted PDF and anonymous supplemen-
98 tary artifacts are content-addressed as immutable evidence objects: each payload leaf is identified
99 by SHA-256 and included in a canonical manifest; verification recomputes leaf and manifest hashes
100 offline. This establishes byte identity and provenance closure, not biological correctness, therapeutic
101 efficacy, or correctness of every scientific inference.

102 8 Conclusion

103 Hash-valid custody is necessary but insufficient for trustworthy evidence pipelines. Explicit termi-
104 nal states and invariants surface failure modes that remain syntactically admissible—a prerequisite
105 before any biomedical performance claim.

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